辐射换热

哈尔滨工业大学

3.辐射换热

3.1 基本概念

3.2 物理、数学模型

3.3 方程及演绎

3.4 原理及应用

3.1 基本概念

- 辐射机理及特点

机理:物体高于绝对零度向外发射光子。特点：不需要直接接触或媒介；伴随热能与辐射能的转化。

- 黑体、白体、透明体

黑体：辐射能全部被物体吸收（A=1）。

白体：投射辐射能全部被反射（R=1）。

透明体：投射辐射能全部透过（D=1）。

- 黑体辐射能

E^0（$W \cdot m^{-2}$）黑体单位表面积，单位时间辐射的能量。

3.1 基本概念

➢ 黑体辐射定律

- **布朗克定律**

辐射强度I_λ^0（$W{\cdot}m^{-2}{\cdot}\mu m^{-1}$）*物体单位面积在单位时间内，单位波长间隔的辐射能量。*

$$I_\lambda^0 = \frac{C_1\lambda^{-5}}{\exp\left(\frac{C_2}{\lambda T}\right) - 1}$$

C1=3.742×10^{8}W•μm^4•m^{-2};C2=3.439×10^4μm•K

- **维恩位移定律**

$$\lambda_m T = 2898\mu m \cdot K$$

- **斯蒂芬定律**

$$E^0 = \int_0^\infty I_\lambda^0 \, \mathbf{d}\lambda = \sigma_0 T^4$$

σ_0=5.67×10^{-8}W•m^{-2}•K-4斯蒂芬-玻尔兹曼常数

- **克希霍夫定律**

$$\varepsilon = A$$

3.1 基本概念

➢ 角系数与互换性定理

Φij：表面i辐射出的能量投射到表面j的比例。

$$\varphi_{ij}F_i = \varphi_{ji}F_j$$

➢ 灰体

灰体的辐射能与实际物体相同，辐射光谱形状与黑体相同。

➢ 实际物体的黑度(发射率)

实际物体辐射能与同一温度的黑体辐射能之比。

$$\varepsilon = \frac{E}{E^0}$$

➢ 灰体有效辐射力

J(W·m^{-2})单位时间内物体单位面积上辐射的总能量。

$$J = E + RG$$

3.1 基本概念

- **投射能**

G（$W \cdot m^{-2}$）单位时间投落到物体单位面积上的总辐射能。

- **隔热屏**

显著降低辐射热流的平板。

- **薄壁构件冷却过程能量守恒定律**

金属冷却dt时间传给冷却介质的热量dQ=金属降低dT引起自身的热焓减少量。

$$\mathbf{d}Q = -C_p M\,\mathbf{d}T$$

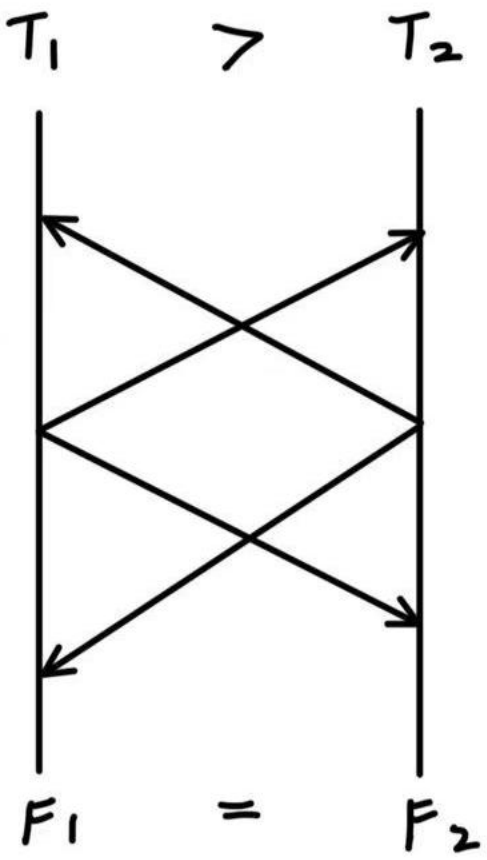
- **对流+辐射换热综合交换系数**

$$h_\Sigma = \frac{J_\Sigma}{T\infty - T_S}$$

3.2 物理、数学模型

- 模型
  - 黑体
    - 两张无限大平板
    - 球体包围
  - 实际物体
    - 两张无限大平板
    - 球体包围
      - FI=F2
      - F1<<F2
    - 圆筒向外部环境
    - 圆筒通过隔热屏向外部环境
    - 综合传热

黑体

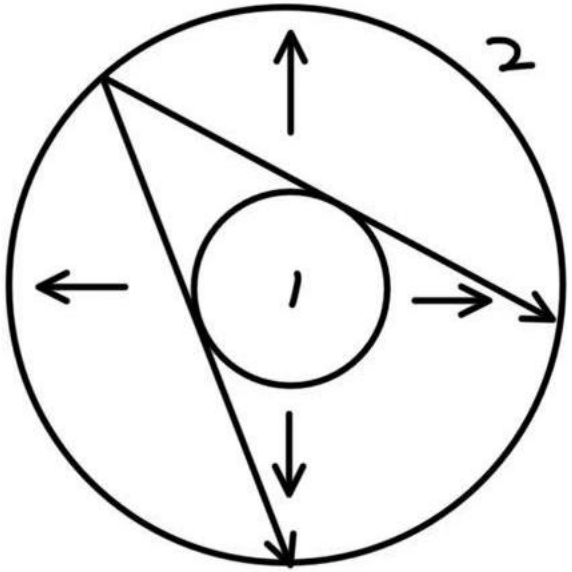


$$Q_{12}^{\circ} = Q_1^{\circ} - Q_2^{\circ}$$

$$= E_1^{\circ} F_1 - E_2^{\circ} F_2$$

$$= C_0 \left[\left(\frac{T_1}{100}\right)^4 - \left(\frac{T_2}{100}\right)^4 \right] F \ (W)$$

黑体



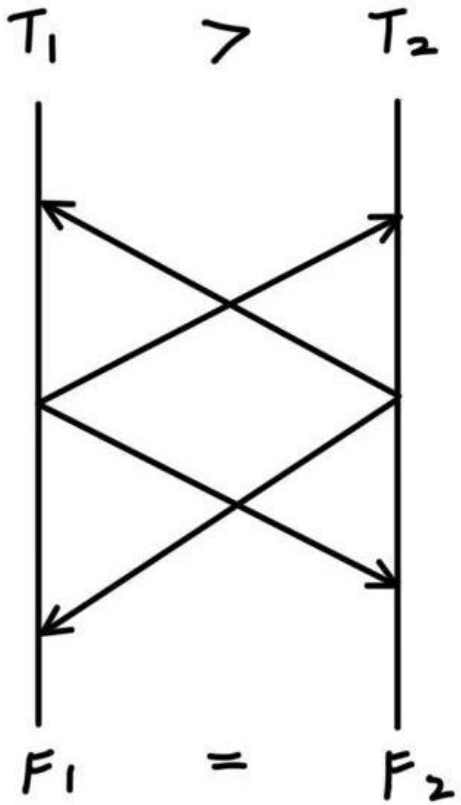
$$\varphi_{12} = 1$$

$$F_1 \varphi_{12} = F_2 \varphi_{21}$$

$$Q_{12}^{0} = E_1^{0} F_1 \varphi_{12} - E_2^{0} F_2 \varphi_{21}$$

$$= C_0 \left[\left(\frac{T_1}{100}\right)^4 - \left(\frac{T_2}{100}\right)^4 \right] F_1 \varphi_{12}$$

实际物体



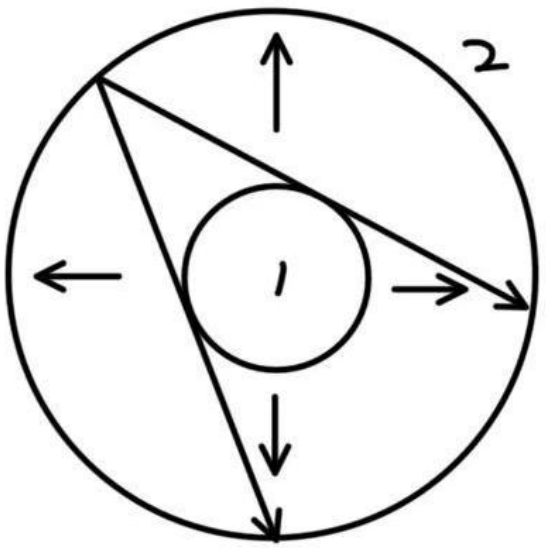
无隔热屏

$$Q_{12} = \frac{E_1^0 - E_2^0}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1} F$$

有隔热屏

$$Q_{12} = \frac{E_1^0 - E_2^0}{\left(\frac{1}{\varepsilon_1} - \frac{1}{2}\right) + \left(\frac{1}{\varepsilon} - \frac{1}{2}\right) + \cdots + \left(\frac{1}{\varepsilon} - \frac{1}{2}\right) + \left(\frac{1}{\varepsilon_2} - \frac{1}{2}\right)} F$$

实际物体



$$\varphi_{12} = 1 \qquad \varphi_{21} = \frac{F_1}{F_2}$$

$$Q_{12} = \frac{\sigma_0 (T_1^4 - T_2^4)}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1} + \frac{1}{F_1 \varphi_{12}} + \frac{1-\varepsilon_2}{\varepsilon_2 F_2}}$$

$$= \frac{F_1 \sigma_0 (T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1} + \frac{F_1}{F_2}\left(\frac{1}{\varepsilon_2} - 1\right)}$$

$$= \begin{cases} F_1 \varepsilon_1 \sigma_0 (T_1^4 - T_2^4) & F_1 \ll F_2 \\ \dfrac{F_1 \sigma_0 (T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1} & F_1 \approx F_2 \end{cases}$$

实际物体

热量传递路径

圆筒(F_1) ⟶ 辐射 ⟶ 外部环境(F_∞)

薄壁圆筒降温速率

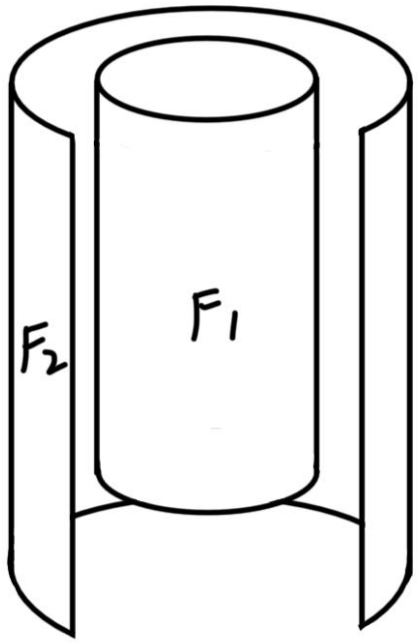
能量守恒：内能下降率 = 筒辐射热流量

$$-mc_p\frac{dT}{dt}\Big|_{1\infty} = F_1\varepsilon_1\sigma_0(T_1^4 - T_\infty^4)$$

$$\int_0^t dt = -\frac{\rho\delta c_p}{\varepsilon_1\sigma_0}\int_{T_0}^{T}\frac{dT}{T_1^4 - T_\infty^4}$$

实际物体

$$F_1 \xrightarrow{\text{辐射}} F_2\,(\text{隔热屏}) \xrightarrow{\text{辐射}} F_\infty$$



内能降低速率 $= Q_{12} = Q_{2\infty}$

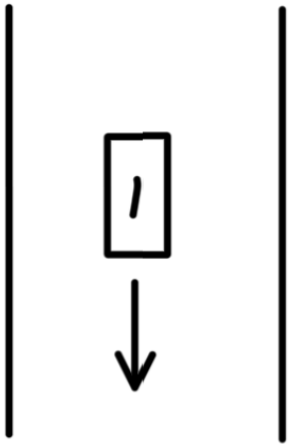
$$-mc_p\frac{dT}{dt}\Big|_{1s\infty} = Q_{12} = Q_{2\infty}$$

$$Q_{12} = \frac{F_1\sigma_0(T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1} \qquad F_1 \approx F_2$$

$$Q_{2\infty} = F_2\varepsilon_2\sigma_0(T_2^4 - T_\infty^4) \qquad F_2 \ll F_\infty$$

$$\begin{cases} \dfrac{dT_1}{dt}\Big|_{1s\infty} = -\dfrac{F_1\sigma_0}{mc_p}\cdot\dfrac{T_1^4 - T_2^4}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1} \\ \varepsilon_2(T_2^4 - T_\infty^4) = \dfrac{T_1^4 - T_2^4}{\frac{1}{\varepsilon_1}+\frac{1}{\varepsilon_2}-1} \end{cases}$$

实际物体



1 对流+辐射换热量

$$Q = Q_c + Q_r$$

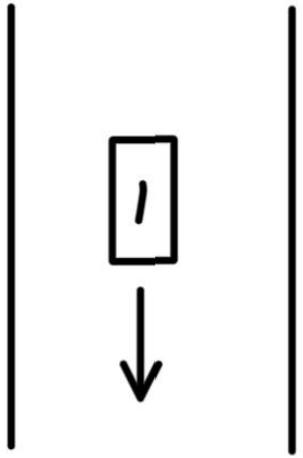
$$= h_c(T-T_\infty)F_1 + \frac{\sigma_0(T^4-T_\infty^4)F_1}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1}+\frac{1}{F_1\varphi_{12}}+\frac{1-\varepsilon_2}{\varepsilon_2 F_2}}\cdot\frac{T-T_\infty}{T-T_\infty}$$

$$= h_c(T-T_\infty)F_1 + h_r(T-T_\infty)F_1$$

$$\therefore Q = h_\Sigma(T-T_\infty)F_1$$

$$h_\Sigma = h_c + h_r = h_c + \frac{\sigma_0}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1}+\frac{1}{F_1\varphi_{12}}+\frac{1-\varepsilon_2}{\varepsilon_2 F_2}}\cdot\frac{T^4-T_\infty^4}{T-T_\infty}$$

实际物体



2、冷却时间

能量守恒：

$$-m c_p \frac{dT}{dt}\Big|_{F_1} = h_\Sigma (T - T_\infty) F_1$$

$$\frac{dT}{dt}\Big|_{F_1} = -\frac{F_1 h_\Sigma}{\rho c_p V}(T - T_\infty)$$

$$\int_0^{t_F} dt = -\frac{\rho c_p V}{F_1}\int_{T_0}^{T} \frac{dT}{h_\Sigma (T - T_\infty)}$$

$$t_F = -\frac{\rho c_p V}{F_1}\int_{T_0}^{T} \frac{dT}{h_\Sigma (T - T_\infty)}$$

3.3 方程及演绎

$$Q_{12} = J_1 F_1 \varphi_{12} - J_2 F_2 \varphi_{21}$$

$$= \frac{J_1 - J_2}{\frac{1}{F_1 \varphi_{12}}}$$

如果 $T_1 > T_2$，物体1=争失 $Q(-)$，物体2=争得 $Q(+)$

$$J = E + GR = E + (1-A)G = E + (1-\varepsilon)G = \varepsilon E_0 + (1-\varepsilon)G$$

$$\Rightarrow G = \frac{J - \varepsilon E_0}{1-\varepsilon}$$

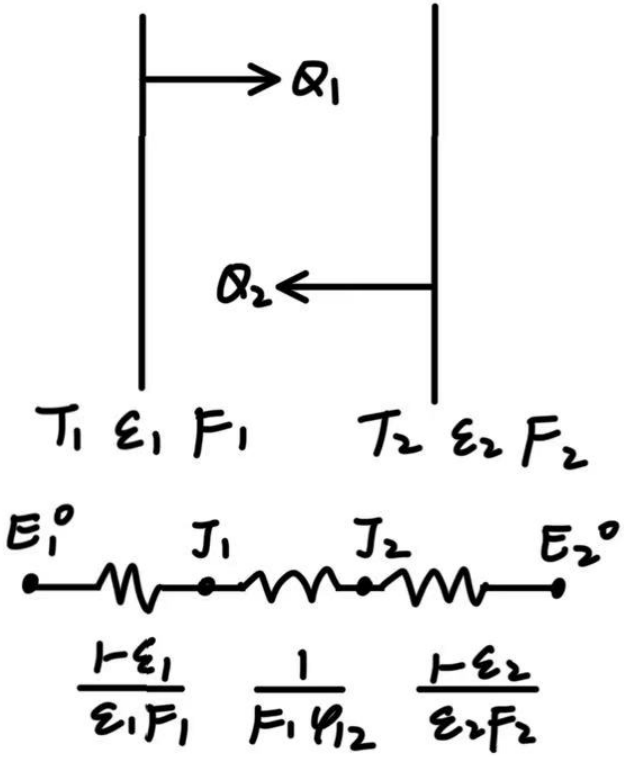
$$Q(-) = (J_1 - G_1)F_1 = \left(J_1 - \frac{J_1 - \varepsilon_1 E_1^0}{1-\varepsilon_1}\right)F_1 = \frac{E_1^0 - J_1}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1}}$$

$$Q(+) = (G_2 - J_2)F_2 = \left(J_2 - \frac{J_2 - \varepsilon_2 E_2^0}{1-\varepsilon_2}\right)F_2 = \frac{J_2 - E_2^0}{\frac{1-\varepsilon_2}{\varepsilon_2 F_2}}$$

$$Q_{12} = Q(-) = Q(+)$$

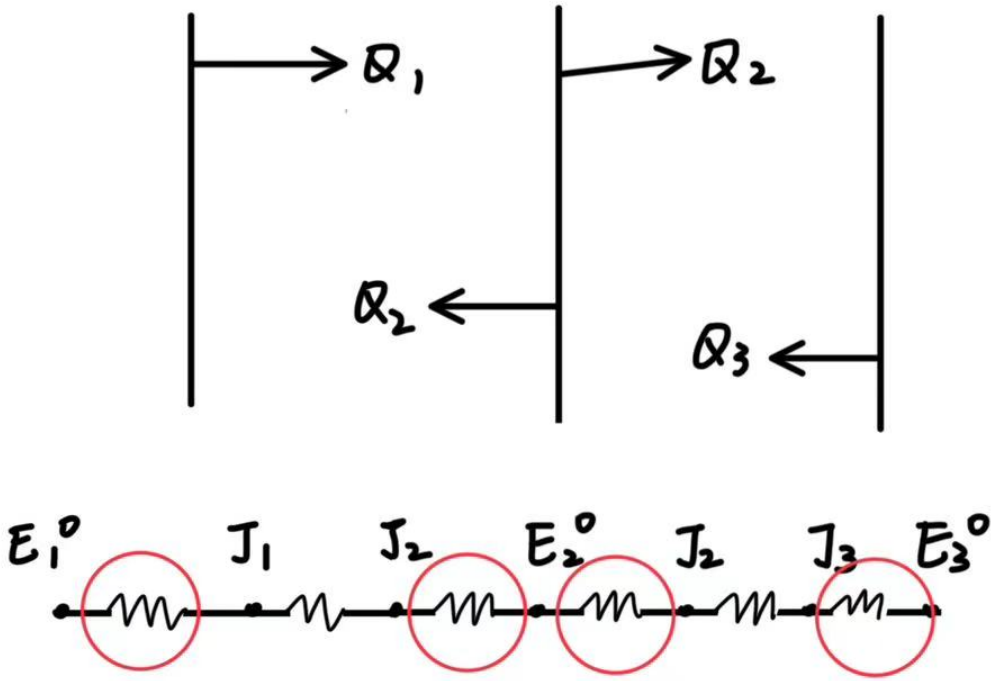
$$Q_{12} = \frac{E_1^0 - E_2^0}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1} + \frac{1}{F_1 \varphi_{12}} + \frac{1-\varepsilon_2}{\varepsilon_2 F_2}}$$

3.3 方程及演绎



表面辐射内阻：$\frac{1-\varepsilon}{\varepsilon F}$

表面辐射外阻：$\frac{1}{F_i\varphi_{ij}}$



"4个内阻"

$Q_{13} = \frac{1}{2} Q_{12}$

3.4 原理及应用

四、作业：高温真空炉设计基础

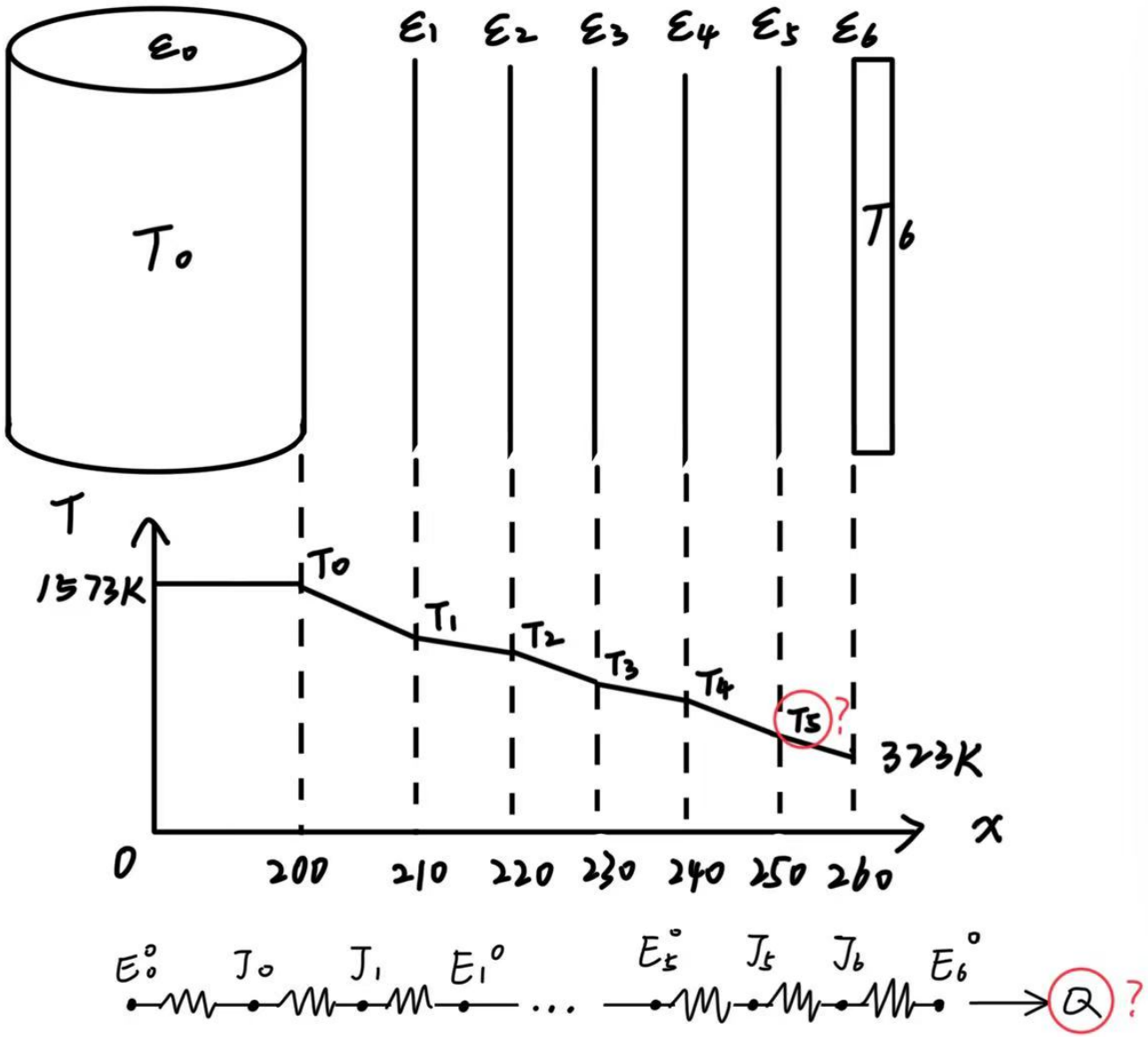
设某高温真空炉用管状石墨电热体、五层金属隔热屏和水冷外壳结构。求石墨电热体经隔热屏向炉壳的辐射传热量及第五层隔热屏的温度。

已知：电热体的温度T_0=1573K，水冷外壳内壁温度T_6=323K；

石墨电热体、隔热屏1-5及水冷外壳内壁的直径分别为400、420、440、460、480、500、520，高度均为800；

石墨电热体、隔热屏1-5及水冷外壳内壁的黑度分别为ε_0、ε_1、ε_2、ε_3、ε_4、ε_5和ε_6。

3.4 原理及应用



真空炉 ⇨ 只有辐射

3.4 原理及应用

$$Q_{01} = \frac{F_0\sigma_0(T_0^4 - T_1^4)}{\frac{1}{\varepsilon_0} + \frac{F_0}{F_1}(\frac{1}{\varepsilon_1} - 1)}$$ 令 $R_{01} = \frac{1}{\varepsilon_0} + \frac{F_0}{F_1}(\frac{1}{\varepsilon_1} - 1)$

则 $Q_{01} = \frac{F_0\sigma_0(T_0^4 - T_1^4)}{R_{01}}$

同理 $Q_{12} = \frac{F_1\sigma_0(T_1^4 - T_2^4)}{R_{12}}$, $R_{12} = \frac{1}{\varepsilon_1} + \frac{F_1}{F_2}(\frac{1}{\varepsilon_2} - 1)$

$Q_{23} = \frac{F_2\sigma_0(T_2^4 - T_3^4)}{R_{23}}$, $R_{23} = \frac{1}{\varepsilon_2} + \frac{F_2}{F_3}(\frac{1}{\varepsilon_3} - 1)$

…

稳态时 $Q_{01} = Q_{12} = Q_{23} = \cdots = Q$

$$Q = \frac{F_0\sigma_0(T_0^4 - T_6^4)}{R_{总}}$$

所以 $\frac{R_{总}}{F_0} = \frac{R_{01}}{F_0} + \frac{R_{12}}{F_1} + \frac{R_{23}}{F_2} + \frac{R_{34}}{F_3} + \frac{R_{45}}{F_4} + \frac{R_{56}}{F_5}$

$$\boxed{Q = \frac{\sigma_0(T_0^4 - T_6^4)}{\sum_{K=0}^{5} \frac{R_{K\,K+1}}{F_K}}}$$

3.4 原理及应用

$$Q = \frac{F_5 \sigma_0 (T_5^4 - T_6^4)}{R_{56}}$$

$$\boxed{T_5^4 = \frac{R_{56} Q}{F_5 \sigma_0} + T_6^4}$$